LB-AO: Response to the Technical Review

What we accept, what we defend, and how the critique changes our development plan

Musculoskeletal Digital Twin | Independent Study
Prepared for Corbin Wright and collaborators | October 7, 2026

Overall position

The review is useful and directionally strong. Its most consequential criticism is that our software does not yet enforce the distinctions and contracts established by our design. We should preserve the conceptual architecture while strengthening integrity checks and testing it through a simulation consumer.

The revised sequence is **integrity → minimal vertical slice → full planar squat → conditions → personalization**. The next milestone should demonstrate that an evidence-backed datum can reach an equation of motion, change an observable output, and remain traceable to its source and assumptions.

What deserves preservation

Evidence and assumptions remain separate. Anatomical claims, measured quantities, estimated parameters, biomechanical conventions, and modeling assumptions should not silently become interchangeable.

Property roles remain explicit. A parameter such as optimal fiber length differs from a state-dependent function such as moment arm r(q), and from an output such as muscle force F(t).

Unresolved values remain honest gaps. Model-specific parameters should stay unresolved until a formulation and defensible resolution method have been chosen. Symmetry remains an overridable assumption; uncertainty and transformation history must survive conversion.

Scope and evidence status

This response evaluates LB-AO_Technical_Review.pdf and consolidates our preceding discussion. The review reports execution of 18 tests and eight malformed-data probes. Those results are attributed to the reviewer; this document does not claim an independent rerun of the repository or an independent verification of the cited anatomical literature. Proposed changes are decisions for future implementation, not completed fixes.

Next action: strengthen the existing gluteus maximus template and test it through dynamics before propagating it to many more muscles. The critique changes our sequencing while reinforcing the evidence architecture that makes this project distinctive.

Reference [R] LB-AO_Technical_Review.pdf, “Technical Review of LB-AO: Critique of current work and a proposed improvement plan,” prepared for Corbin Wright and collaborators, October 7, 2026, 8 pages. Page references identify review claims; the original primary studies were not independently re-audited for this response.

1. Findings to adopt

Enforce contracts between layers

The reported coordinate mismatch (HIP_FLEXION_R versus HIP_FLEX_EXT_R), pennation-property mismatch, and class-name incompatibilities are substantive integrity problems [R, pp. 3-4]. Fixing individual names is necessary, but insufficient. Every target, source, coordinate, frame, property, and derivation dependency must resolve under explicit cross-module rules.

Class reconciliation should use declared hierarchy or explicit mappings where appropriate. Similar names do not automatically establish equivalence, and a schema need not assign quantitative properties to every anatomical class. The critical test is whether intended properties reach their intended entities.

Validate invalid data, not only valid examples

The reviewer reports that all eight malformed records were accepted, despite 18 existing tests passing [R, pp. 7-8]. This is the strongest software finding: successful positive tests do not demonstrate enforcement. Add negative tests for malformed enums, null required units, physically invalid values, unresolved IDs, and unsupported assumptions, with actionable error messages.

Validation must be conditional on datum status and property semantics. An unresolved datum need not carry a populated numeric value. An assumed datum must carry a rationale and the appropriate provenance; it should not receive an automatic exemption from scrutiny.

Preserve claim-specific evidence and applicability

Evidence should support a particular relationship or quantitative claim, not merely an entity as a whole. Cohort metadata must not leak from a shared default into unrelated model-derived or unresolved records [R, p. 4]. Distinguish evidence-population applicability from intended simulation applicability, and record how one informs the other.

Use one defaults-resolution implementation and test its precedence. Defaults are a convenience, not a license to copy evidential context across records. Explicit subject-specific and side-specific overrides must remain possible.

Make property definitions authoritative

Reconcile role/form inconsistencies for musculotendon length and muscle force, then check each datum against its property definition [R, p. 3]. Bilateral records must express compatible coordinate contexts on both sides, or be split into side-specific records. Shared assumptions must never conceal a right-only coordinate reference.

2. Recommendations to refine

A declarative schema complements semantic checks

Adopt typed structural validation, for example with Pydantic, while retaining dedicated semantic and evidence checks. The useful sequence is **structural validation → semantic integrity → evidence/provenance validity → simulation readiness**. A library for types and enums cannot by itself decide whether a derivation is justified or a model has all required inputs.

LinkML remains an option when interoperability or schema generation makes it useful. It is not a prerequisite for the next milestone. Keep these validation responsibilities explicit without requiring four separate software services.

Canonical facts and derived inverse relationships

The 73 PART_OF edges without stored HAS_PART inverses are not automatically 73 missing anatomical facts [R, p. 4]. Store a canonical direction and derive its inverse in queries where the semantics guarantee it. Document this behavior and test it. Two independently mutable copies of one fact create avoidable opportunities for disagreement.

Unused vocabulary is not inherently defective

A controlled vocabulary may define predicates before the dataset uses them. The requirement is that every relationship actually used in data or documentation has defined semantics, domain/range rules where needed, inverse behavior, and versioning. Reconcile undocumented predicates; do not invent anatomical edges merely to use every term.

Compatible units versus incorrect relabeling

Appendix A treats changing a kg record to lb as a rejection case. Pounds can be a valid mass unit if accepted by policy and converted correctly. A validator can check dimensional compatibility; it generally cannot infer that a numeric value was incorrectly relabeled without conversion history or source context. Test both valid conversion and known relabeling errors. An incompatible unit or a missing required unit should fail.

Different values require reconciliation, not automatic conflict

The reported PCSA values, 33.4 cm^2 and 30.37 cm^2, warrant explicit comparison [R, p. 4]. Before labeling them conflicting evidence, determine whether they represent the same muscle aggregation, normalization, population, and intended quantity. They may be alternative measured and model-adapted estimates. Record the difference, selection rationale, applicability, and downstream effect; mark a conflict when comparability justifies it.

3. Scientific and scope defenses

Anatomical correctness and resolution have different priorities

Correct false or misleading anatomical claims now. Add detailed compartments, landmarks, and attachment surfaces when a downstream model needs that resolution. The review flags adductor magnus innervation, gluteal attachments, and incomplete biarticular topology [R, pp. 4-5]; these are prompts to verify the underlying sources, not a reason to accept every proposed correction without checking.

Biarticular topology becomes essential when introducing the knee. Deferred origins and insertions should already name their missing module and reason. More detailed gluteus maximus attachment surfaces may improve anatomical fidelity without being required by the first sagittal hip demonstration.

Defer neurological detail without misrepresenting laterality

We can defer duplicating every nerve into sided entities for a mechanics-first model. However, a single nerve node must be identified as a generic anatomical type or an explicitly scoped abstraction; it must not imply that one physical nerve innervates both sides. Laterality metadata alone does not resolve the type-versus-instance ambiguity. Use clear left, right, bilateral, midline, unspecified, or not-applicable semantics, and block unsupported laterality-dependent analyses.

Mean and SD do not identify a distribution

The reported pennation mean of 22° and SD of 26° caution against unconstrained normal sampling [R, p. 6]. They do not establish a log-normal distribution. Preserve the reported summary statistics and leave the distribution unknown unless supported by evidence. If sampling is needed, record the chosen distribution, physical support, fitting method, and rationale as a modeling assumption.

Truncating a normal distribution or fitting a log-normal can change the resulting moments. Check the actual sampled statistics and bounds; do not imply that the sampled distribution automatically reproduces the published mean and SD.

Version layers independently, release them together

Different ontology, schema, dataset, and document version numbers are not inherently incompatible. A single project release should identify the exact component versions it contains. Add a release manifest, changelog, compatibility policy, and migration guidance. Preserve stable identifiers or supply aliases and explicit migrations when correcting names. This carries forward our commitment to backward compatibility.

Keep the first simulation deliberately small

An ankle-knee-hip planar squat is a valuable next model. Before it, a minimal hip/segment demonstration can test the full data-to-dynamics path with fewer simultaneous unknowns. It is an architectural experiment, not evidence of physiological or clinical validity.

4. Development commitments

Phase A - Integrity

Reproduce the reported defects, fix confirmed D1-D8 issues with the refinements above, and introduce structural, semantic, and provenance checks. Establish canonical relations, explicit mappings, defaults precedence, unit rules, claim-level evidence, and a compatible release manifest. Integrate the checks into continuous integration.

Exit criterion: confirmed malformed cases fail clearly; legitimate conversions and unresolved records pass under documented rules; cross-file references resolve; baseline data and supported legacy inputs retain their intended meaning.

Phase B - Minimal vertical slice

Resolve a gluteus maximus parameter from YAML into a small sagittal hip/segment model. Explicitly supply segment inertia, geometry, initial conditions, and the chosen force-to-torque abstraction, with provenance for every required input. Test a parameter perturbation and its measurable effect on trajectory or torque.

Exit criterion: a selected input change produces an interpretable output change, the equation and integrator pass basic numerical checks, and the run records the exact values, units, assumptions, and dataset versions used.

Phase C - Full planar squat

Extend to ankle, knee, hip, and trunk/segment representation only as the task requires. Define contacts or support constraints, coordinate conventions, actuation, and a specific muscle formulation before resolving its missing parameters. Validate mechanics and numerical behavior before adding conditions.

Phases D and E - Conditions and personalization

Represent conditions as justified parameter modifications with applicability and uncertainty. Start with changes that the model can actually express and observe; for example, a sagittal model alone cannot support a full analysis of frontal-plane hip abductor weakness. Add sensitivity analysis using explicitly chosen sampling assumptions.

Then connect measurements to parameters through defined observation models and identifiability analysis. Apply subject-specific > side-specific > shared-baseline precedence, report uncertainty, and keep unresolved parameters distinguishable from estimated ones.

A core invariant: what number did the simulation use, and why?

The reported derivation, 30.37 cm^2 × 61 N/cm^2 = 1852.57 N, rounded to 1852.6 N, is a useful template [R, pp. 3, 8]. Every resolved parameter should retain input IDs, source context, assumptions, conversions, derivation, uncertainty treatment, and resolution choice. A provenance viewer should expose that chain, not reconstruct it after a run.